

Topics on Electrical Engineering

B.Sc

Drive Technology

LectureSlide

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1. Introduction
2. Magnetic Circuits and Materials

Introduction



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Content Preview



- The goal of this lecture is to give an introduction to the fascinating world of electric machines.
- We will start from understanding how they work to the types currently used in commercial and industrial applications.
- The lecture will focus on mathematical and engineering concepts which will have direct use with real-life engineering applications.
- This lecture is a total of 3 SWS with a total of forty-five (45) UE.
- A unit (UE) is defined as 45 min lecture.



- For two (2) sessions there will be a self-guided study which will require you to work through the material on your own.
- In these sessions you should focus on two topics: **Electromechanic Energy conversion** and **Rotating Magnetic Fields**. You can either access this content on WebBook or LectureBook which hold the **same** content.
- The next face-to-face session will focus on a recap of these topics along with any questions you may have. Afterwards there will be a 15 min long quiz covering these two topics **only**.



- Lecture materials and all possible supplements will be present in its Github Repo.
 - You can easily access the link to the web-page from [here](#).

Github is chosen for easy access to material management and CI/CD capabilities and allowing hosting websites.

- In the lecture content is also distributed as a WebBook which can be accessed from the [Repo website](#).



- The student should have a good background in circuit analysis and familiarity with integral/differential calculus.
- After completion, the student will have a good grasp of electrical machines from their construction, how they are designed, what types out there and what applications are possible.

Requirements	Taught Lecture	Code	Degree	Outcome
Vector Calculus	Mathematik 1	MAT	B.Sc	Python Programming
Differential Calculus	Mathematik 2	MAT	B.Sc	Electric Drive Fundamentals
Linear Algebra	Mathematik 1	MAT	B.Sc	Magnetic Circuitis
Physics	Physik	PHY		Electric Motor Analysis
-				-

Table 1: Distribution of materials across the semester.



Description	Value
Official Name	Antriebstechnik
Lecture Code	ANT
Module Code	MECH-B-5-ANT-ANT-ILV
Lecture Name	Drive Technology
Semester	5
Season	WS
Lecturer	Daniel T. McGuiness, Ph.D
Module Responsible	DaM
Software	Python, Simulink
SWS Total	3
UE Total	45
ECTS	5
Working Language	English



- The lecture will have a single personal assignment comprising of a set list of questions and a final exam comprising of all the topics covered in the lecture. Instructions will be given in the assignment.
- For the written exam you are allowed to write your personal equation reference paper, as long as it is a single sheet of A4, double sided and contains no exercise or solutions. Your reference sheet will be collected after the exam.

Assignment Type	Value
Personal Assignment	20
Final Exam	30
Lab Work	50
Sum	100



Title
Advanced electric drives: analysis control and modeling using MATLAB/Simulink
Analysis of electric machinery and drive systems
Design of rotating electrical machines
Electric Machinery Fundamentals (5th Edition)
Electric Machinery
Electric Motors and Drives: Fundamentals Types and Applications
Electrical Machines and Drives: Fundamentals and Advanced Modelling
Electrical machines, drives, and power systems

Table 2: Lecture sources which can be useful during the course of the lecture. For more information on sources, please consult the [repo](#).



Topic	Units	Self Study
Magnetic Circuits and Materials	4	4
Transformers	4	4
Electromechanical Energy Conversion	4	8
Lab 1: Installation of a Drive System	3	6
Rotating Magnetic Fields	2	4
Lab 2: Building a Lathe	4	8
Synchronous Machines	4	8
Poly-phase Induction Drives	4	8
Lab 3: Synchronous and Induction Drives	4	8



Topic	Units	Self Study
DC Drives	4	8
Special Machines	2	3
Lab 4: Photovoltaic Cells	4	8
Final Exam	2	4
Total	45	90

Magnetic Circuits and Materials



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Maxwell's Equations

Flux Linkage, Inductance and Energy

Magnetic Materials

Introduction

Ferromagnetism

B-H Curve

AC Excitation

Definitions

Hysteresis Losses

Bibliography

List of References



- (LO1) An Overview of Maxwell's Equations,
- (LO2) Introduction to Magnetic Circuits,
- (LO3) Brief look on Magnetic Materials,
- (LO4) Magnetic Material Losses.





- Almost all electric drives use **ferromagnetic material** for shaping and directing **B** -fields.
 - **B** -fields act as the medium for **transferring/****converting** energy.

Permanent Magnet (PM)s are also widely used in drive design.

- Without these materials, practical implementations of most familiar electro-mechanical devices would not be possible.
- Analysing and describing systems containing them is essential for designing effective drives.
- We start by looking at Maxwell's equations.



- Maxwell's equations are a set of coupled partial differential equations which form the foundation of electric and magnetic circuits.
- In their PDE form, they are written as [1]:

$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\varepsilon_0}, & \nabla \cdot \mathbf{B} &= 0, \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t}, & \nabla \times \mathbf{B} &= \mu_0 \left(\mathbf{J} + \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right).\end{aligned}$$

where $\mathbf{E}$ is the electric field, $\mathbf{B}$ is the magnetic field, ρ is the electric charge density and $\mathbf{J}$ the current density, ε_0 is the vacuum permittivity and μ_0 is the vacuum permeability.



- Solution to Maxwell's equation can be hard if not attainable and usually simplifying assumptions are made to reach practical solutions.
- First, the displacement-current term ($\epsilon_0 \partial \mathbf{E} / \partial t$) can be neglected.

This term accounts for **B**-fields produced in space by time-varying **E**-fields (i.e., electromagnetic radiation) [2].

- Neglecting $\epsilon_0 \partial \mathbf{E} / \partial t$ results in the magneto-static form which relates **B**-fields to the currents (**J**) which produce them.
- In integral form [3]: 6 / 56

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \int_S \mathbf{J} \cdot d\mathbf{a}, \quad \text{and} \quad \oint_S \mathbf{B} \cdot d\mathbf{a} = 0.$$



$$\oint_{\mathcal{C}} \mathbf{H} \cdot d\mathbf{l} = \int_{\mathcal{S}} \mathbf{J} \cdot d\mathbf{a}, \quad \text{and} \quad \oint_{\mathcal{S}} \mathbf{B} \cdot d\mathbf{a} = 0.$$

The line integral ($\oint$) of the tangential component of the magnetic field intensity ($\mathbf{H}$) around a closed contour ($\mathcal{C}$) is equal to the total current ($\mathbf{J}$) passing through any surface ($\mathcal{S}$) linking that contour.

- The second one states the magnetic flux density ($\mathbf{B}$) is conserved, i.e., no net flux enters or leaves a closed surface (i.e., $\nabla \cdot \mathbf{B} = 0$).

These allow us to remove $\mathbf{E}$ field from our calculations.



- A second simplifying assumption involves the concept of **magnetic circuits**.
- The general solution for the **H** and the **B** in a structure of complex geometry is extremely difficult if not practically impossible.
- However, a 3D field problem can often be reduced to what is essentially a circuit equivalent of magnetic elements,
 - with acceptable engineering accuracy [4].



- A magnetic circuit is a structure composed of **highly permeable**,

$$\mu_r \gg 0$$

material(s).

- The presence of high-permeability causes magnetic flux to be **confined** to the paths defined by the structure.

This is similar to how currents are confined to the conductors of an electric circuit.

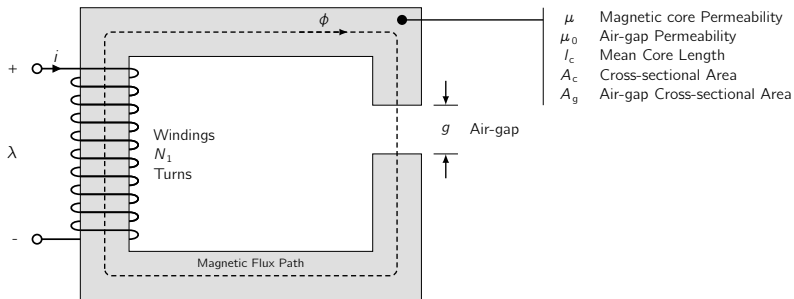


Figure 1: An example of a magnetic circuit.



Let's have a further look at this setup:

- The core is assumed to be composed of magnetic material with permeability much greater than of surrounding air (i.e., $\mu_r \gg \mu_0$).
 - The core is of **uniform cross section** and is excited by a winding of N turns carrying a current of i amperes.
-
- This winding produces a **B** -field in the core.



- Due to high permeability of the material (i.e., $\mu_r \gg \mu_0$), the magnetic flux is confined **almost entirely to the core**,
- The ***B***-field lines follow the path defined by the core,
- The flux density is essentially uniform over a cross section as the cross-sectional area is **uniform**.
- The ***B***-field can be visualised in terms of flux lines which form closed loops interlinked with the winding.



- The source of the $\mathbf{B}$ -field in the core is the ampere-turn Ni .
- In magnetic circuit terminology Ni is the Magneto-motive Force (MMF) acting on the magnetic circuit.
- In systems with more than one winding, Ni must be replaced by the **algebraic sum** of the ampere-turns of all the windings:

$$\sum_{n=0}^{\infty} N_i i_i.$$

- The magnetic flux (ϕ) crossing a surface $\mathcal{S}$ is the surface integral of the normal component of $\mathbf{B}$:

$$\phi = \int_{\mathcal{S}} \mathbf{B} \cdot d\mathbf{a}$$

where the unit of flux (ϕ) is Wb.



$$\phi = \int_S \mathbf{B} \cdot d\mathbf{a}$$

- This equation states that all flux entering the surface enclosing a volume must leave the volume as they form closed loops.
- This is enough to justify the uniformity of the magnetic flux density ($\mathbf{B}$) across the cross section of a magnetic circuit (A_c):

$$\phi_c = B_c A_c,$$

where ϕ_c is the flux, B_c is the magnetic flux density magnitude and A_c is the cross-sectional core area.



- Using the original magneto-static simplifications (i.e., no $\mathbf{E}$ -fields), we can come up to:

$$\mathcal{F} = Ni = \oint \mathbf{H} d\mathbf{l}.$$

- As the path of the flux line is close to the mean length¹ of the core (l_c), this is simplified to:

$$\mathcal{F} = Ni = H_c l_c,$$

where H_c is the **average magnitude** of $\mathbf{H}$.



- The relationship between the $\mathbf{H}$ and $\mathbf{B}$ is a **property of the material**.
- It is common to assume a linear relationship shown as:

$$\mathbf{B} = \mu \mathbf{H},$$

where μ is the magnetic permeability.

- In SI units, $\mathbf{H}$ is measured in $\text{A} \cdot \text{m}^{-1}$ and $\mathbf{B}$ is in $\text{Wb} \cdot \text{m}^{-2}$ or T.



- Transformers are wound on **closed cores**.
- However, electro-mechanical devices which incorporate a **moving element** must have **air gaps** in their magnetic circuits.
- When the air-gap (g) is much smaller than the dimensions of the adjacent core faces, the magnetic flux will follow the path defined by the core and the air gap and the techniques of magnetic-circuit analysis can be used.

If g becomes excessively large, the flux will be leak out of the sides of the air gap and the techniques of magnetic-circuit analysis will no longer be strictly applicable.

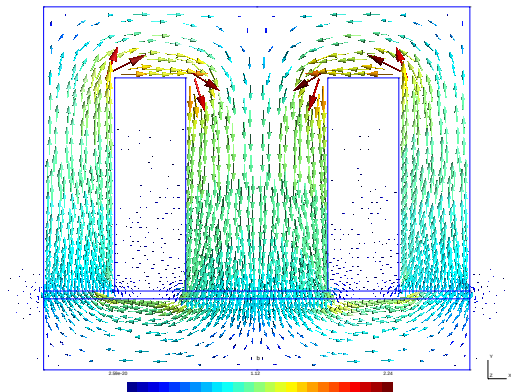


Figure 2: A Magnetic Circuits with two mmf sources.

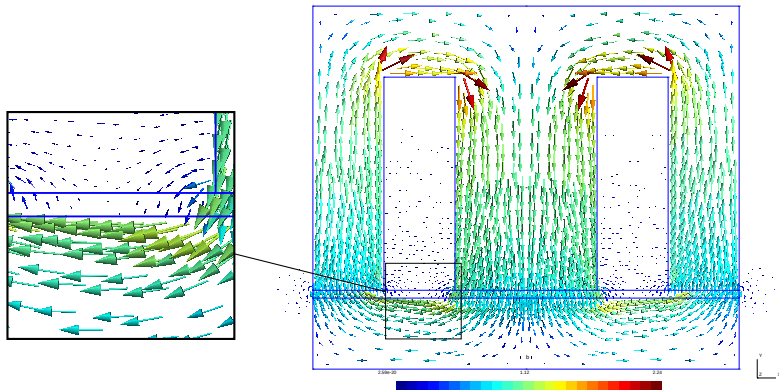


Figure 3: A closeup of the fringe paths caused by the air gap.



- Provided the air-gap length g is sufficiently small, the configuration can be analysed as a magnetic circuit with two (2) series components:
 1. Magnetic core (μ), cross-sectional area A_c , and mean length l_c ,
 2. Air gap (μ_0), cross-sectional area A_g , and length l_g .
- In the core the flux density can be assumed uniform and ($\mu \gg \mu_0$):

$$B_c = \frac{\phi}{A_c}, \quad \text{and in the air gap,} \quad B_g = \frac{\phi}{A_g},$$



- Applying this to the magnetic circuit gives:

$$\mathcal{F} = H_c l_c + H_g g,$$

using the ***B-H*** relationship:

$$\mathcal{F} = \frac{B_c}{\mu} l_c + \frac{B_g}{\mu_0} g.$$

- Here, $\mathcal{F}$ is the MMF applied to the magnetic circuit where $\mathcal{F}_c = H_c l_c$ is the magnetic field in the core and $\mathcal{F}_g = H_g l_g$ is the magnetic field in the air-gap.
- Using ϕ we can re-write this equation as:

$$\mathcal{F} = \phi \left(\frac{l_c}{\mu A_c} + \frac{g}{\mu_0 A_g} \right).$$



- The term which multiplies mmf to flux is called **Reluctance**.

$$\mathcal{F} = \phi \left(\frac{\overbrace{I_c}^{\mathcal{R}_c}}{\mu A_c} + \frac{\overbrace{g}^{\mathcal{R}_g}}{\mu_0 A_g} \right) \quad \text{or} \quad \mathcal{F} = \phi (\mathcal{R}_c + \mathcal{R}_g)$$

Reluctance can be seen as analogous to resistance to electrical circuits.



- When a **B**-field varies with time, an **E**-field is produced in space as determined by Faraday's law [1]:

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$
$$\oint_C \mathbf{E} \cdot d\mathbf{s} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}.$$

The line integral of the electric field intensity **E** around a closed contour $\mathcal{C}$ is equal to the time rate of change of the magnetic flux linking (i.e. passing through) that contour.

In electric drives, the effects of **E** field can be neglected.



- This simplification allows us to cancel the LHS of the equation to just the induced voltage (i.e., electromotive force)

$$\oint_C \mathbf{E} \cdot d\mathbf{s} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}.$$

- Whereas on the RHS, the flux is dominated by the core flux (ϕ).

$$\mathcal{E} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}.$$



- As winding links the core flux N times, our equation is simplified to:

$$\mathcal{E} = N \frac{d\phi}{dt} = \frac{d\lambda}{dt} \quad \text{where} \quad \lambda = N\phi \quad \blacksquare$$

where λ is flux linkage, measured in (Wb · Turn),



- For a magnetic circuit composed of magnetic material of constant magnetic permeability, the relationship between ϕ and i will be **linear** and we can define the inductance L as:

$$L = \frac{\lambda}{i} \quad \text{which can be written as} \quad L = \frac{N^2}{\mathcal{R}_{\text{tot}}}$$

where L is the inductance measured in Henry (H).

Inductance requires a linear relationship between flux and mmf.

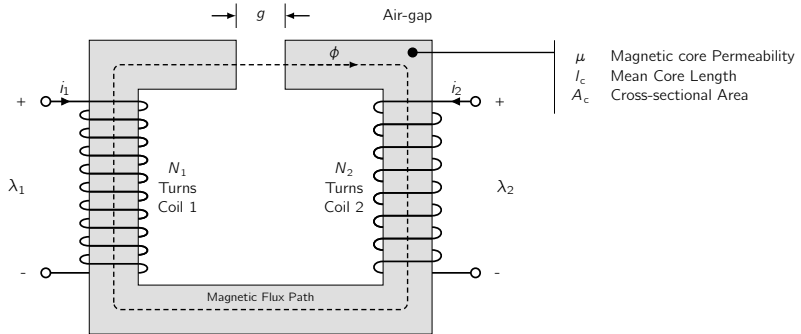


Figure 4: An example of a magnetic winding circuit with 2 windings.



- The MMF ($\mathcal{F}$) is given by the total ampere-turns.
- The reference directions for the currents (i_1, i_2) have been chosen to produce flux in the same direction.
- The total mmf is therefore:

$$\mathcal{F} = N_1 i_1 + N_2 i_2.$$

- With the reluctance of the core neglected and assuming that $A_c = A_g$, the core flux ϕ is:

$$\phi = (N_1 i_1 + N_2 i_2) \frac{\mu_0 A_c}{g},$$

where ϕ is the **resultant core flux** produced by the total mmf of the two windings.



- This resultant ϕ determines the operating point of the core material.
- If ϕ is broken up into terms attributable to the individual currents, the resultant flux linkages of coil 1 can be expressed as:

$$\lambda_1 = N_1 \phi = \overbrace{N_1^2 \left(\frac{\mu_0 A_c}{g} \right)}^{L_{11}} i_1 + \overbrace{N_1 N_2 \left(\frac{\mu_0 A_c}{g} \right)}^{L_{12}} i_2,$$

and this is simplified to:

$$\lambda_1 = L_{11} i_1 + L_{12} i_2,$$

where L_{11} is the **self** inductance and L_{12} is the **mutual** inductance of the coil.



- Similarly, the flux linkage on coil 2 is:

$$\lambda_2 = N_2 \phi = \overbrace{N_1 N_2 \left(\frac{\mu_0 A_c}{g} \right)}^{L_{21}} i_1 + \overbrace{N_1^2 \left(\frac{\mu_0 A_c}{g} \right)}^{L_{22}} i_2$$

and this is simplified to:

$$\lambda_1 = L_{21} i_1 + L_{22} i_2$$

where $L_{21} = L_{12}$ is mutual inductance and L_{22} is self inductance.

It is important to note the ϕ is calculated based on the superposition of i_1 , i_2 which implies a linear relationship between flux and mmf.



- It cannot be rigorously applied in situations where the **nonlinear characteristics of magnetic materials** dominate.
- However, in many situations of practical interest, the reluctance of the system is dominated by that of an air gap (which is of course linear) and the nonlinear effects of the magnetic material can be ignored.
- In other cases it may be perfectly acceptable to assume an average value of magnetic permeability for the core material.
- Using this it is possible to calculate a corresponding average inductance which can be used for calculations of reasonable engineering accuracy.



- Electro-mechanical devices require two (2) important properties:
 - Obtain large magnetic flux densities ($\uparrow \mathbf{B}$)
 - Requiring relatively low levels of magnetizing force ($\downarrow \mathbf{H}$).

Since magnetic forces and energy density increase with increasing flux density, this effect plays a large role in the performance of energy-conversion devices.



- Magnetic materials can be used to **constrain and direct B -fields** in well-defined paths.
- In a transformer, they are used to maximize the coupling between the windings as well as to **lower the excitation current** required for transformer operation.
- In electric drives, they are used to shape the fields to obtain desired torque-production and electrical terminal characteristics.

- Alkali Metal
- Alkaline Earth Metal
- Metal
- Metalloid
- Non-metal
- Halogen
- Noble Gas
- Lanthanide/Actinide



- Typically composed of iron (**Fe**) and alloys of iron with:
 - Cobalt, Tungsten, Nickel, Aluminium, and other metals
- These materials are characterised by a wide range of properties,
 - Phenomena responsible for their properties **are the same**.
- Ferro-magnetic materials are found to be composed of a large number of **domains**.
 - Regions where the magnetic moments of all the atoms are parallel, giving rise to a net magnetic moment for that domain.
- In an un-magnetised sample of material, the domain magnetic moments are **randomly** oriented, and the net resulting magnetic flux in the material is zero (**0**).

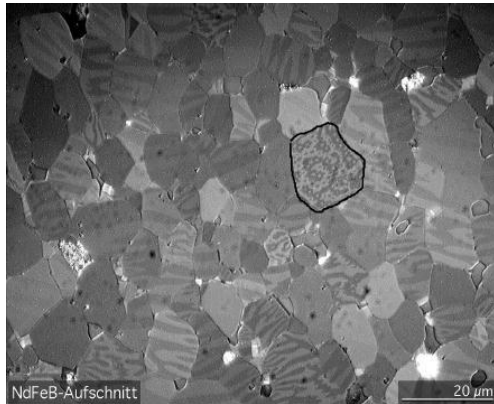


Figure 6: Microcrystalline grains within a piece of Nd₂Fe₁₄B (the alloy used in neodymium magnets) with magnetic domains made visible with a Kerr microscope. The domains are the light and dark stripes visible within each grain. The outlined grain has its magnetocrystalline axis almost vertical, so the domains are seen end-on [5].



- When **B** -field is applied to the material, the magnetic domain moments tend to **align** with the applied **B** -field.
- As a result, the magnetic moments add to the **B** -field, producing a higher flux density than would exist due to the magnetising force alone.
- Therefore the effective permeability μ , equal to the ratio of the total magnetic flux density to the applied magnetic-field intensity, is large compared with the permeability of free space (μ_0).
- As the magnetising force (**H**) is increased, this continues until all the magnetic moments are aligned with the applied field.
 - at this point they can no longer contribute to increasing the magnetic flux density

Material becomes fully saturated.



- Without an external $\mathbf{B}$ -field, magnetic domains naturally align along certain directions associated with the **crystal structure** of the domain.
 - This is known as axes of easy magnetisation [6].
- If the applied $\mathbf{B}$ is reduced, the domain magnetic moments relax to the direction of easy magnetism nearest to the applied field.
- As a result, $\mathbf{B}$ is reduced to zero (0), although they will tend to relax towards their initial orientation, the magnetic dipole moments will no longer be totally random in their orientation;
- They will retain a net magnetisation **along the applied** field.
- It is this effect which is responsible for the phenomenon known as magnetic hysteresis [7].

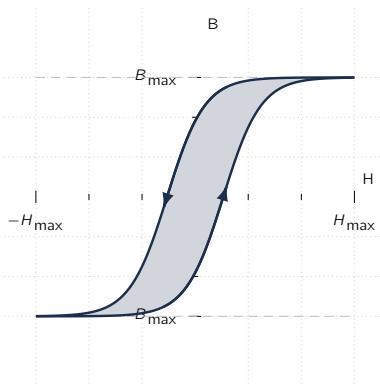


Figure 7: An example of an B-H Hysteresis curve.



1st Quadrant

- The domains are very small, below the single domain size where there is resistance to demagnetization [8].
- The domains start to increase in size.

For a small interval, the magnetization will be reversible.

- As the field increases, magnetization will no longer reverse to zero but move on a minor hysteresis loop.
- Eventually the curve starts to bend over to the right.
 - It will still increase as more magnetic domains reach their full size and their magnetisation become parallel to the external field [9].



1st Quadrant - Saturation

- Eventually the **B** -field will become high enough where no more change in the magnetisation occurs.
 - This is called **technical saturation**.
- It is possible to reach 99+% saturation.
- As the field is backed off from “saturation”, the magnetisation declines very slightly to the B_r point.
 - This is the remanence, or residual induction [10].

All of the magnetic energy is now in the magnet and its field.



2nd Quadrant - Demagnetisation

- In this quadrant the applied field **opposes** the materials **B** -field.
- As the external **B** -field increases in magnitude, some domains will reverse.
- At the knee of the demagnetisation curve, this increase has become rapid and the magnetisation will fall to the H_{ci} point.
- At H_{ci} , the number of domains aligned with the original magnetization is the same as the number aligned with the opposing magnetic field.

The net magnetisation is zero (0).



3rd Quadrant - Re-magnetisation

- The total magnetization of the part will be **reversed**.
- If we go far enough, magnetization will reach the saturation level in the negative direction.

4th Quadrant - Demagnetisation

- After fully reversing the magnet and removing the field in the third quadrant, magnetization will recoil to a point that is the negative of the B_r observed when in the first and second quadrant.
- If we apply additional field in the positive direction, we duplicate the second quadrant curve.



- In AC power systems, voltage and flux wave-forms closely approximate sinusoidal functions over time.
- Time to describe the excitation characteristics and losses associated with steady-state ac operation of magnetic materials.
- Assume a closed-core magnetic circuit (i.e., with no air gap),
- And a sinusoidal variation of the core flux (ϕ) with the following:

$$\phi(t) = \phi_{\max} \sin(\omega t) = A_c B_{\max} \sin(\omega t)$$

where:

$\phi_{\max}$ amplitude of core flux in Wb,

$B_{\max}$ Amplitude of flux density B_c in T,

ω Angular frequency in rad/s,

f Frequency in Hz.



- From, the voltage induced in the N -turn winding is:

$$e(t) = \omega N \phi_{\max} \cos(\omega t) = E_{\max} \cos(\omega t)$$

where:

$$E_{\max} = \omega N \phi_{\max} = 2\pi f N A_c B_{\max}$$

- In steady-state AC, it is more important to use rms rather than instant values.
- Generally, the rms value of a periodic function of time $f(t)$, of T is:

$$F_{\text{rms}} = \sqrt{\left(\frac{1}{T} \int_0^T f^2(t) dt \right)}$$



- The rms value of a sine wave can be shown to be $1/\sqrt{2}$ times its peak value. Therefore the induced voltage rms is:

$$E_{\text{rms}} = \frac{2\pi}{\sqrt{2}} f N A_c B_{\text{max}} = \sqrt{2} \pi N A_c B_{\text{max}} = 4.44 N A_c B_{\text{max}} \quad \blacksquare$$

- To produce magnetic flux in the core requires current in the exciting winding known as the exciting current, i_ϕ ,

The nonlinear magnetic properties of the core require that the waveform of the exciting current differs from the sinusoidal waveform of the flux.

- When the hysteresis curve saturates the excitation current spikes.

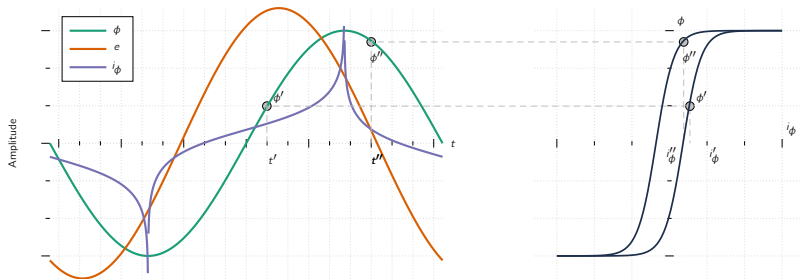


Figure 8: An Example Visualisation of the excitation current experienced by a magnetic circuit.



- The exciting current supplies the mmf required to produce the core flux and the power input associated with the energy in the magnetic field in the core.
- Part of this energy is dissipated as losses and results in heating of the core.
- The rest appears as reactive power associated with energy storage in the magnetic field.

This reactive power is not dissipated in the core as it is cyclically supplied and absorbed by the excitation source.



- There are two (2) mechanisms associated with time varying $\mathbf{B}$ -fields.

Ohmic Heating

- Associated with induced currents in the core material.
- From Faraday's law, we see that $\mathbf{B}$ -fields give rise to $\mathbf{E}$ -fields.
- In magnetic materials these $\mathbf{E}$ -fields result in induced currents,
 - These are known as eddy currents, circulating in the core material, opposing the material's internal $\mathbf{B}$ -field [11].



- To counteract the corresponding demagnetizing effect by the **eddy currents**, the current in the exciting winding must increase.

Therefore the resultant "dynamic" B-H loop under ac operation is somewhat "fatter" than the hysteresis loop for slowly varying conditions, and this effect increases as the excitation frequency is increased.

- It is for this reason that the characteristics of electrical steels vary with frequency and hence manufacturers typically supply characteristics over the expected operating frequency range of a particular electrical steel.



- To reduce the effects of eddy currents, magnetic structures are usually built of **thin sheets of laminations** of the magnetic material.
- These laminations, which are aligned in the direction of the field lines, are insulated from each other by an oxide layer on their surfaces or by a thin coat of insulating enamel or varnish.
- This greatly reduces the magnitude of the eddy currents since the layers of insulation interrupt the current paths; the thinner the laminations, the lower the losses.

In general, eddy-current loss tends to increase as the square of the excitation frequency and also as the square of the peak flux density.



- Due to the **hysteretic nature** of magnetic material.
- In a magnetic circuit or the transformer, a time-varying excitation (i_ϕ) will cause the magnetic material to undergo a cyclic variation described by a hysteresis loop.
- The energy input W to the magnetic core as the material undergoes a single cycle is shown to be:

$$W = \oint i_\phi d\lambda = \oint \left(\frac{H_c l_c}{N} \right) (A_c N dB_c) = A_c l_c \oint H_c dB_c$$

- Notice $A_c l_c$ is the **core volume** and the integral is the area of the ac hysteresis loop, we see that each time the magnetic material undergoes a cycle, there is a net energy input into the material.



- This is the required energy to move around the magnetic dipoles in the material and is dissipated as heat in the material.
- Therefore for a given flux level, the corresponding hysteresis losses are **proportional to the area of the hysteresis loop** and to the **total volume of material**.
- As there is an energy loss per cycle, hysteresis power loss is **proportional to the frequency of the applied excitation**. ([Data Sheet](#))



- In general, both losses depend on the metallurgy of the material as well as the flux density and frequency.
- Information on core loss is typically presented in graphical form.
- It is plotted in terms of watts per unit weight as a function of flux density;
 - Often a family of curves for different frequencies are given.
 - Generally it is either 50 or 60 Hz.



- Nearly all transformers and certain sections of electric machines use sheet-steel material that has highly favorable directions of magnetization along which the core loss is low and the permeability is high.
- This material is termed **grain-oriented steel** [12].
- The reason lies in the atomic structure of a crystal of the silicon-iron alloy,
 - which is a body-centred cube.

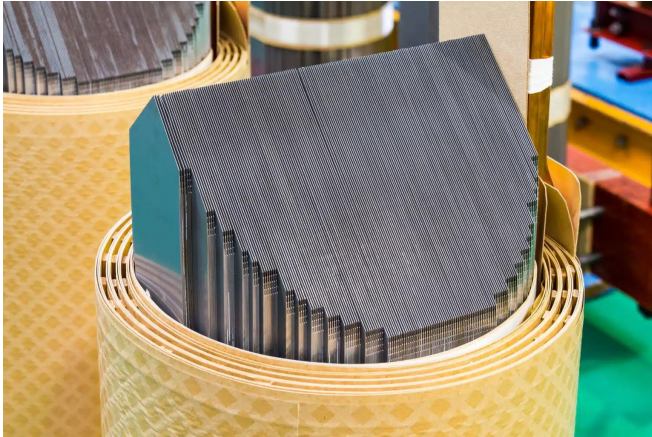


Figure 9: An example of Grain Oriented Steel used in industry [13].



MMF The magnetic pressure which drives magnetic flux in a magnetic circuit, and is the work done to move a unit magnetic pole.. 26, 34, 41

PM An object made from a material that is magnetized and creates its own persistent magnetic field. 17



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